

Additional file

Early-life Exposure to Three Size-fractionated Ultrafine and Fine Atmospheric Particulates in Beijing Exacerbates Asthma

Development in Mature Mice

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Additional file 1

Figure S1. mRNA expression in mouse lungs was detected 2 days after

exposure to PM by OA. The gene expression of *GATA3*, a Th2 transcription factor, was elevated. In contrast, the mRNA expression of *MyD88* was decreased following PM exposure. * $p < 0.05$ vs. Con group (n = 4-5 mice/group).

Figure S2. PM induces oxidative stress in vitro and in vivo. (A) The

increased expression of heme oxygenase (HO-1) in THP-1 cells induced by PM was determined. (B) The effects of PM on HO-1 expression in RAW 264.7 cells. Cells were treated with PM (10, 20, 50 $\mu\text{g/ml}$) for 5 h. RAW 264.7 cells were incubated with 20 mM NAC for 1 h before addition of 25 $\mu\text{g/ml}$ PM for 5 h. (C) HO-1 expression in lung tissues was measured 48 h after the last exposure to PM or PBS. Induction of HO-1 expression by PM exposure was observed in vivo. n = 3-5 mice/group. The data are presented as the mean $\pm$ SEM. * $p < 0.05$ vs. Con, # $p < 0.05$ vs. 25 $\mu\text{g/ml}$ PM without NAC, & $p < 0.05$ vs. OVA/PM (F1, F2 or F3).

Table S1. Primers used for quantitative real time PCR analysis.

Table S2. The parameters involved in inflammatory responses with statistical difference between three size-fractionated ultrafine and fine atmospheric particulate matter groups in OVA-induced asthma model.

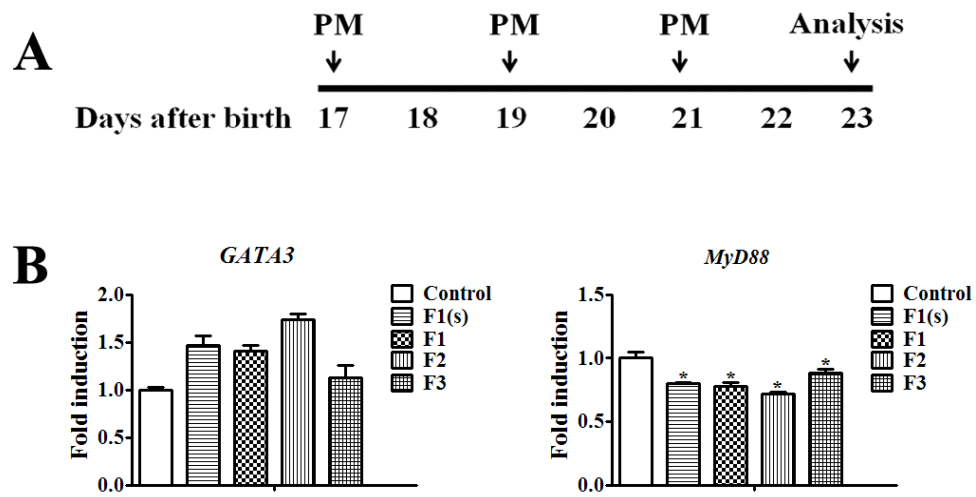


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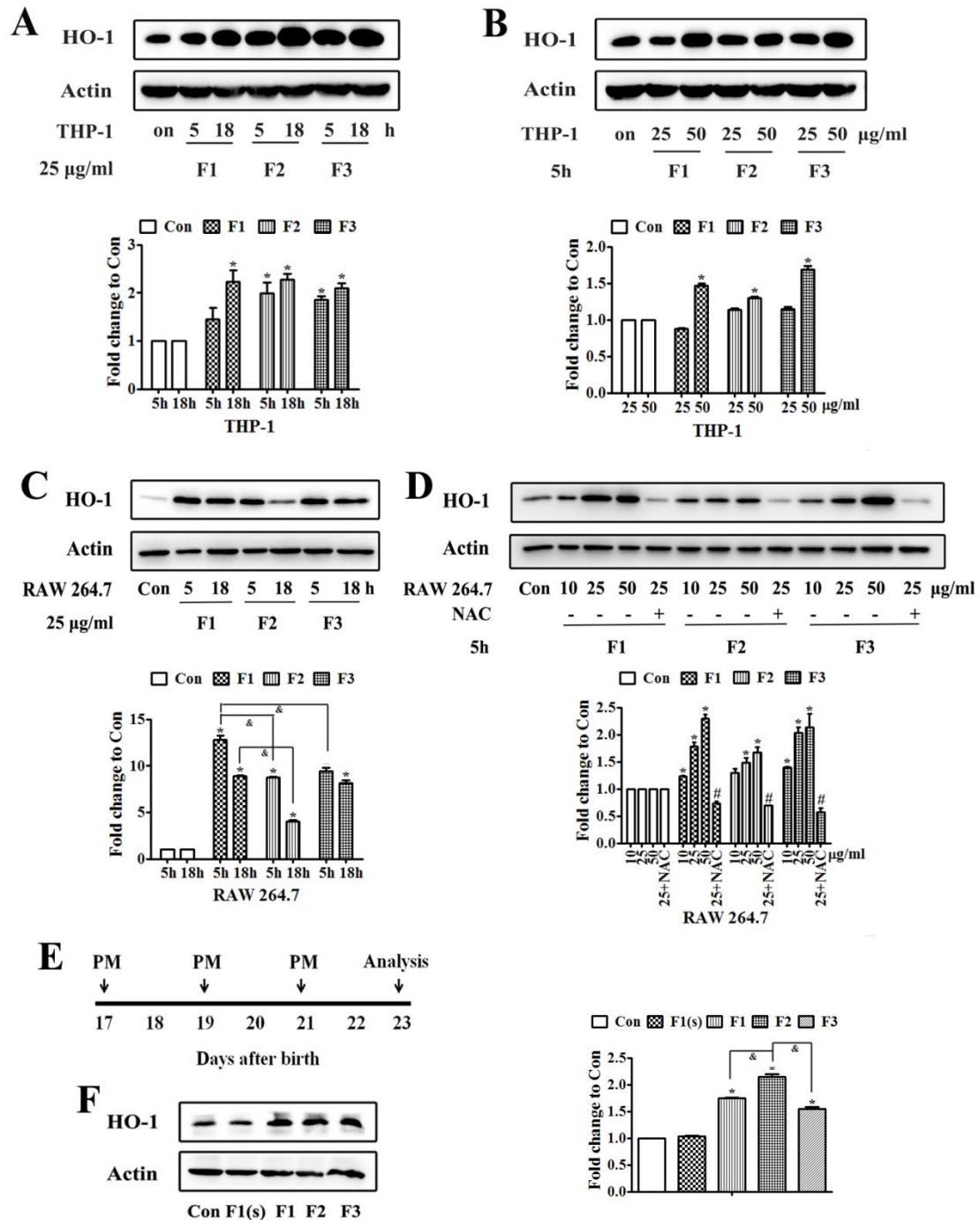


Figure S2. PM induces oxidative stress in vitro and in vivo. (A) The increased expression of heme oxygenase (HO-1) in THP-1 cells induced by PM was determined. (B) The effects of PM on HO-1 expression in RAW 264.7 cells. Cells were treated with PMs (10, 20, 50 µg/ml) for 5 h. RAW 264.7 cells were incubated with 20 mM NAC for 1 h before addition of 25 µg/ml PMs for 5 h. (C) HO-1 expression in lung tissues were measured at 48 h after the last exposure to PMs or PBS. Induction of HO-1 expression by PM exposure was observed in vivo. n=3-5 mice/group. The

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Table S1. Primers used for quantitative real time PCR analysis

Gene	Primer	Sequence
Mouse GAPDH	Forward	GACTTCAACAGCAACTCCCAC
	Reverse	TCCACCACCCTGTTGCTGTA
Mouse Dnmt1	Forward	GTCGGACAGTGACACCCTTT
	Reverse	TTTAGTGGGGCCCTTCGTG
Mouse Dnmt3b	Forward	CCTCCGACGCAGGAA AGATT
	Reverse	ACTTGGGTGGCTCCTGAAAT
Mouse IL-4	Forward	GCATGGAGTTTTTCCCATGTT
	Reverse	AGATGGATGTGCCAAACGTC
Mouse IL-5	Forward	CTCTGTTGACAAGCAATGAGACG
	Reverse	TCTTCAGTATGTCTAGCCCCTG
Mouse IL-13	Forward	CCTCTGACCCTTAAGGAGCTTAT
	Reverse	CGTTGCACAGGGGAGTCT
Mouse IL-33	Forward	ATTTCCTCCGGCAAAGTTCAG
	Reverse	AACGGAGTCTCATGCAGTAGA
Mouse IL-10	Forward	CAGAGCCACATGCTCCTAGA
	Reverse	GTCCAGCTGGTCCTTTGTTT
Mouse Ccl11/Eotaxin	Forward	TTCTATTCTGCTGCTCACGG
	Reverse	AGGGTGCATCTGTTGTTGGTG
Mouse Ccl3/MIP-1α	Forward	ACTGCCTGCTGCTTCTCCTACA
	Reverse	AGGAAAATGACACCTGGCTGG
Mouse CXCL1/KC	Forward	CTGGGATTACCTCAAGAACATC
	Reverse	CAGGGTCAAGGCAAGCCTC
Mouse CXCL2	Forward	TGTCCTCAACGGAAGAACC
	Reverse	CTCAGACAGCGAGGCACATC
Mouse Muc5ac	Forward	CTGTGACATTATCCCATAGCCC
	Reverse	AAGGGGTATAGCTGGCCTGA
Mouse Muc5b	Forward	CATGGATGGCTGCTTCTGT
	Reverse	TAAATTCGGCCACCGTGT
Mouse GATA3	Forward	CCCATTACCACCTATCCGCC
	Reverse	GTTACACACTCCCTGCCTT
Mouse MyD88	Forward	CCACTCGCAGTTTGTGGAT
	Reverse	TCCTCACGGTCTAACAAGGC

Table S2 The parameters involved in inflammatory responses with statistical difference between three size-fractionated ultrafine and fine atmospheric particulate matter groups in OVA-induced asthma model

Relevant parameters involved in asthma inflammation ^a	OVA/PM groups that have significant difference between them ^a
HO-1 expression in RAW264.7 cells (5 h)	OVA/F1 vs OVA/F2, OVA/F1 vs OVA/F3
HO-1 expression in RAW264.7 cells (18 h)	OVA/F1 vs OVA/F2, OVA/F3 vs OVA/F2
HO-1 expression in lung tissues	OVA/F2 vs OVA/F1, OVA/F2 vs OVA/F1
<i>Dnmt1</i> expression in RAW264.7 cells	OVA/F1 vs OVA/F2, OVA/F1 vs OVA/F3
<i>Dnmt3b</i> expression in RAW264.7 cells	OVA/F1 vs OVA/F2, OVA/F1 vs OVA/F3
<i>Dnmt3b</i> expression in lung tissues	OVA/F3 vs OVA/F1, OVA/F3 vs OVA/F1
Lymphocyte in BALF	OVA/F1 vs OVA/F2, OVA/F1 vs OVA/F3
Neutrophil in BALF	OVA/F1 vs OVA/F2, OVA/F1 vs OVA/F3
Eosinophil in BALF	OVA/F1 vs OVA/F3, OVA/F2 vs OVA/F3
Total IgE of serum	OVA/F2 vs OVA/F1, OVA/F2 vs OVA/F1
<i>Muc5ac</i> expression in lung tissues	OVA/F2 vs OVA/F1, OVA/F2 vs OVA/F1
<i>Muc5b</i> expression in lung tissues	OVA/F1 vs OVA/F3, OVA/F2 vs OVA/F3
PAS staining	OVA/F1 vs OVA/F3, OVA/F2 vs OVA/F3
<i>IL-4</i> expression in lung tissues	OVA/F1 vs OVA/F2, OVA/F1 vs OVA/F3
<i>IL-5</i> expression in lung tissues	OVA/F1 vs OVA/F3, OVA/F2 vs OVA/F3
<i>IL-13</i> expression in lung tissues	OVA/F1 vs OVA/F2, OVA/F1 vs OVA/F3
<i>IL-33</i> expression in lung tissues	OVA/F1 vs OVA/F2, OVA/F1 vs OVA/F3
<i>IL-10</i> expression in lung tissues	OVA/F1 vs OVA/F2, OVA/F1 vs OVA/F3
<i>CXCL1/KC</i> expression in lung tissues	OVA/F1 vs OVA/F3, OVA/F2 vs OVA/F3

^a Refers to statistical difference analysis of relevant parameters involved in inflammatory responses between OVA and PM groups. The OVA/PM group(s) that have significant difference between them ($p < 0.05$) is (are) listed in the table.